

Clinton Power Station
8401 Power Road
Clinton, IL 61727



U-604399

10 CFR 50.73
SRRS 5A.108

December 18, 2018
U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Clinton Power Station, Unit 1
Facility Operating License No. NPF-62
NRC Docket No. 50-461

Subject: Licensee Event Report 2017-008-01

Enclosed is Licensee Event Report (LER) 2017-008-01: Division 3 Shutdown Service Water Pump Start Failure. This is the supplemental report to LER 2017-008-00 submitted to the NRC on August 11, 2017. The updated information in the LER is denoted by revision bars located in the right-hand margin. This report is being submitted in accordance with the requirements of 10 CFR 50.73.

There are no regulatory commitments contained in this report.

Should you have any questions concerning this report, please contact Mr. Dale Shelton, Regulatory Assurance Manager, at (217) 937-2800.

Respectfully,

A handwritten signature in black ink, appearing to read "Theodore R. Stoner".

Theodore R. Stoner
Site Vice President
Clinton Power Station

KP/lam

Attachment: Licensee Event Report 2017-008-01

cc:

Regional Administrator – Region III
NRC Senior Resident Inspector — Clinton Power Station
Office of Nuclear Facility Safety — Illinois Emergency Management Agency

IEZZ
NRR



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name

Clinton Power Station, Unit 1

2. Docket Number

05000461

3. Page

1 OF 5

4. Title

Division 3 Shutdown Service Water Pump Start Failure

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
06	15	2017	2017	- 008	- 01	12	18	2018	Facility Name	Docket Number
										05000
										05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
097	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)		

12. Licensee Contact for this LER

Licensee Contact

Mr. Dale Shelton, Regulatory Assurance Manager

Telephone Number (Include Area Code)

(217) 937-2800

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
B	BI	P	Sulzer	Y					

14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

15. Expected Submission Date

Month	Day	Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On June 15, 2017, Clinton Power Station (CPS) commenced procedure CPS 9069.01, Shutdown Service Water Operability Test. The purpose of this procedure is to verify operability of the Division 3 Shutdown Service Water (SX) System Pump 1SX01PC and selected valves per the Inservice Testing program on a quarterly basis. At 0958, SX pump 1SX01PC was started and after approximately 30 seconds, it tripped due to thermal overload. The pump was declared inoperable and Operations entered Technical Specifications (TS) Limiting Condition for Operation (LCO) 3.7.2, Condition A which requires the High Pressure Core Spray (HPCS) system to be declared inoperable and enter TS LCO 3.5.1 Condition B which requires verification by administrative means that the Reactor Core Isolation Cooling (RCIC) system is operable and within 14 days restore the HPCS system to operable status. The cause of the event was an incremental increase in internal pump resistance that could not be overcome by the developed torque of the motor starting from standby conditions. This increased pump resistance was due to unidentified impacts during a design modification (i.e., dry packing at pump start, unidentified corrosion effects on packing shaft sleeve, and inadequate hard-face material). Another root cause was the untimely implementation of a corrective action that resulted from a 2014 causal evaluation due to inadequate procedure guidance. Corrective actions have been completed to restore operability of 1SX01PC and ensure failure mechanisms introduced by design changes have been assessed. This event is reportable under 10 CFR 50.73(a)(2)(v)(D) as a condition that could have prevented the fulfillment of a safety function and 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by Technical Specifications.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Clinton Power Station, Unit 1	05000461	2017	- 008	- 01

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric -- Boiling Water Reactor, 3473 Megawatts Thermal Rated Core Power
Energy Industry Identification System (EIS) codes are identified in text as [XX].

EVENT IDENTIFICATION

Division 3 Shutdown Service Water Pump Start Failure

A. Plant Operating Conditions Before the Event

Unit: 1 Event Date: 6/15/17

Event Time: 0958

Mode: 1 Mode Name: Power Operation

Reactor Power: 97 percent

B. Description of Event

On June 15, 2017, Clinton Power Station (CPS) attempted to perform the quarterly surveillance test of the Division 3 shutdown service water (SX) [BI] pump [P] (1SX01PC). The pump ran for approximately 30 seconds before thermal overloads tripped the pump motor [M].

At 0958, Operators declared the Division 3 SX, Division 3 Emergency Diesel Generator (EDG) [EK], Division 3 Inverter [EF], Division 3 Direct Current Distribution [EJ], Division 3 Alternating Current Distribution [ED], and the High Pressure Core Spray (HPCS) [BG] system inoperable due to the pump trip. Operators entered the Technical Specification Actions Requirements for Limiting Condition for Operation (LCO) 3.7.2 and LCO 3.5.1, ECCS - Operating. Operators verified within one hour that the Reactor Core Isolation Cooling (RCIC) system was operable and entered the action to restore HPCS to operable status within 14 days.

A second pump run was attempted with video recording that showed the motor spinning the shaft more quickly for the first rotation, then immediately slowing down to approximately 70 RPM. The run lasted approximately 8 seconds before the pump was manually secured.

At 2220 hours, Operators completed an 8-hour non-emergency notification to the NRC (Event Notification Number 52806) under 10 CFR 50.72(b)(3)(v)(D) due to the loss of a single train safety system, that is, inoperability of the HPCS system function. This event is reported under 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by Technical Specifications and under 10 CFR 50.73(a)(2)(v)(D) as a condition that could have prevented the fulfillment of a safety function. This is because the Division 3 SX pump was potentially inoperable greater than the 14 days allowed for an inoperable HPCS system by TS 3.5.1 Condition B, (last successful run of the Division 3 SX pump was on March 15, 2017, prior to pump failure on June 15, 2017, and return to service on June 23, 2017). This condition could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

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NARRATIVE**C. Cause of the Event**

The causal evaluation identified two root causes fundamental to the failure of the Division 3 SX Pump:

1. Incremental increase in internal pump resistance that could not be overcome by the developed torque of the motor starting from standby conditions. This increased pump resistance was due to unidentified impacts during a design modification (i.e., dry packing at pump start, unidentified corrosion effects on the packing shaft sleeve, and inadequate hard-face material).
2. The corrective action to replace the Division 3 SX Pump resulting from a 2014 failure to start was not timely to prevent a recurrence. This was due to inadequate procedural guidance for promptly identifying and validating failure modes and effects and developing a remediation strategy for the deficient design condition to support well-timed corrective action in the event that the design condition cannot be immediately corrected.

D. Safety Consequences

The purpose for the Division 3 SX system is to provide cooling water to Division 3 cooling loads including the HPCS pump room coolers [CLR], the Division 3 EDG and the Division 3 Switchgear Heat Removal System (VX). The Division 3 SX header is normally supplied cooling water from the non-safety related Plant Service Water System (WS) [KG]. Under design basis event conditions such as a Loss of Offsite Power (LOOP) or Loss of Coolant Accidents (LOCAs), a Division 3 SX pump start signal is generated and the cross-tie valve [V] (1SX014C) from the WS system closes. At this point the Division 3 SX pump would be providing cooling water flow from the lake to the Division 3 SX cooling loads.

Divisions 1 and 2 safety-related equipment remained fully capable of mitigating the consequences of an accident and were available during the period of this event.

The failure of the Division 3 Shutdown Service Water Pump resulted in inoperability of Division 3 Emergency Diesel Generator safety function and inoperability of the HPCS System safety function.

The event described in this report constitutes a safety system functional failure.

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NARRATIVE**E. Corrective Actions**

Several component changes were made to the overall pump/motor assembly. A new motor with higher starting torque characteristics was installed. The pump breaker and thermal overloads were replaced with new units. The pump shaft sleeves were replaced and the pump packing was replaced with a flexible self-lubricating style equivalent.

On June 23, 2017, following a successful uncoupled run of the new motor, a coupled run was performed with no abnormalities noted. All monitored points showed normal starting characteristics for the pump. The baseline and surveillance test procedures were performed without incident. The Division 3 SX system, including the Division 3 EDG and HPCS system were declared operable at 2345 hours on June 23, 2017.

Additional corrective actions include establishing a preventative maintenance task to periodically inspect the pump packing shaft sleeve for corrosion.

Corrective Actions for the second root cause were to revise plant procedures to include specific guidance to take actions for informing the timeliness of corrective actions for design related issues in cases where the deficient design is still installed or in use and the failure modes and effects of that design are not fully validated. In addition, a design verification of the Division 3 SX pump was completed to ensure failure mechanisms introduced by design changes have been identified. No additional design failure modes were identified.

F. Previous Similar Occurrences

LER 2014-005-01 Failure of Shutdown Service Water Pump Results in Loss of Division 3
Emergency Diesel Generator and High Pressure Core Spray Safety Functions

On September 16, 2014 at approximately 1905 hours, Operators were performing an operability test of the Shutdown Service Water (SX) system. The Division 3 SX pump (1SX01PC) was started to support the 2-year Comprehensive Pump Test and the motor thermal overload protection tripped off the pump after approximately 36 seconds. The cause of the event was that procedures used to develop plant modification change packages contained an inadequate process to identify the need for further reviews and the level of design detail required by those reviews. A new pump was installed and the pump was restored to operable status on September 21, 2014. (ML16006A047).

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NARRATIVE**G. Component Failure Data**

The Division 3 SX pump is a Sulzer Model 8X14A VCM pump with a Siemens-Allis Type RGV motor. The pump is a vertical two-stage deep well pump that consists of a discharge head, six columns, and a bowl assembly to extend 30 feet below the floor elevation. The pump is driven by a 75 horsepower, 1800 RPM, 480-volt motor.